

Aditya Kumar Singh

Senior Software System Designer | 7392898428 | adityakus1996@gmail.com
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SUMMARY

Senior Software Engineer with expertise in **C/C++ system programming, Java backend development**, and cloud-native distributed systems. Strong experience in **Linux internals, multithreading, memory management, concurrency, IPC**, and low-latency networking for 5G-CRAN and CPaaS platforms. Proficient in **Kubernetes, Docker**, and building scalable telecom and cloud systems handling high-concurrency workloads.

EDUCATION

•Indian Institute of Technology (IIT) Kharagpur

Executive Program in Applied AI & Machine Learning

May 2026 – Present

Kharagpur, India (Online)

•Indian Institute of Technology (IIT) Hyderabad

M.Tech in Computer Science Engineering | CGPA: 8.64

2019 – 2022

Hyderabad, India

•University Institute of Engineering & Technology, CSJMU

B.Tech in Computer Science Engineering | CGPA: 7.89

2014 – 2018

Kanpur, U.P.

EXPERIENCE

•AMD

Senior Software System Designer – uProf Performance Profiling & Analysis Tool Development

Sep 2026 – Present

Bengaluru, India

- Contributing to **AMD uProf**, a performance analysis and profiling toolset that helps users understand application and system behavior on AMD platforms.
- Working in systems software with a focus on performance profiling, low-level system behavior, and performance analysis.
- Applying C/C++ and Linux systems knowledge to investigate performance characteristics and support profiling-tool development.

•Radisys India Limited

Senior Software Engineer – Systems & Cloud Infrastructure

Mar 2023 – Sep 2026

Bengaluru, India

CPaaS & SIP Call Control (Java & Cloud)

- **RESTful SIP APIs**: Engineered Java-based REST APIs and integrated the **Cipango SIP stack** via Servlets for real-time call signaling (INVITE, PRACK, BYE).
- **SDP Parsing & Negotiation**: Developed Java utilities to extract and validate media IP/ports via SDP, resolving SIP interoperability issues.
- **Code Modernization & DevOps**: Refactored legacy Java codebases for multi-tenant environments and automated HTTP/HTTPS API deployments using shell scripts.

5G-CRAN Platform (C/C++ & Cloud)

- **DU Feature Development**: Designed and developed DU application features including Multi-FH and Multi-MH connections.
- **Distributed Systems & Kubernetes**: Built a custom **Kubernetes controller in Go** for 5G-RAN profile creation, reducing deployment effort by **30%**.
- **System Performance Tuning**: Tuned Linux CPU governors and investigated NUMA bottlenecks, contributing to a **15% reduction in power consumption**.
- **Advanced Debugging & Stability**: Performed root-cause analysis using **GDB and AddressSanitizer (ASan)** to investigate memory and concurrency issues.
- **CI/CD & Code Quality**: Established a C++ unit testing framework achieving **90%+ code coverage**.

•Dremio Software India Private Limited

Software Engineer – Cloud Platform

Aug 2022 – Feb 2023

Hyderabad, India

- **Billing System Framework**: Designed and developed billing infrastructure for Dremio Cloud using **Java** and **MySQL**.
- **Marketplace Integration**: Integrated billing accounts with **AWS Marketplace** via TACKLE for automated payment workflows.
- **Query Acceleration**: Optimized BI/SQL query performance using Data Reflections, reducing latency by **30%**.
- **Financial Integrity**: Engineered reconciliation logic to detect inconsistent replicas and automate credit reversals.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Golang, Java, Python, MySQL, Bash Scripting
- **Systems & Networking:** Linux Internals, Multithreading, Concurrency, IPC, Memory Management, Low-Latency Networking, Distributed Systems
- **Cloud & DevOps:** Docker, Kubernetes, Git, Maven, CMake, Helm, AWS Marketplace
- **Performance & Debugging:** Performance Profiling, Linux Perf, Intel VTune, GDB, Wireshark, ASan, pidstat, numastat
- **Domain Knowledge:** AMD uProf, 5G-RAN, CPaaS, SIP, SDP

PROJECTS

• Master's Thesis & Intel-Sponsored Research on O-RAN Optimization

Aug 2019 – Jul 2022

IIT Hyderabad

- **O-RAN Optimization:** Performed CAPEX/OPEX analysis for multiple CU-DU-RU split deployment scenarios in 5G O-RAN networks.
- **Open Source Contribution:** Contributed to OpenAirInterface (OAI)-based 5G Cloud-RAN optimization for energy efficiency on Intel Xeon platforms, accelerating L1/L2 workloads using **Intel AVX-2** and **AVX-512** SIMD instructions.
- **Kernel Profiling:** Used Intel VTune, Linux perf, and GDB for kernel and system profiling of performance-critical paths.
- **Automation:** Optimized real-time OAI testbed performance using USRP SDRs and developed Python utilities for log analysis.

RESEARCH AND PUBLICATIONS

- **Conference Paper:** U. Pawar, **Aditya Kumar Singh**, et al., “Understanding Energy Consumption of Cloud Radio Access Networks: an Experimental Study,” *2020 IEEE 3rd 5G World Forum (5GWF)*, 2020. [\[Link\]](#)
- **Poster Presentation:** Shah Alam, **Aditya Kumar Singh**, “Design and Implementation of Image Cryptography using RSA Algorithm,” *IEEE International Conference on Engineering Science & Advance Research*, 2018.

CERTIFICATIONS

- **Architecting with Google Kubernetes Engine: Foundations** – Coursera
- **Data Reflections** – Dremio Software
- **Dremio Fundamentals** – Dremio Software

MISCELLANEOUS

- **Competitive Programming:** Active problem solver on LeetCode, HackerRank, and GeeksforGeeks, focusing on Data Structures & Algorithms and system design concepts.
- **Interests:** Cricket, Chess, Fitness Training